Revision notes responding to professor comments

This page is added for side-by-side review. The original thesis pages follow after this review sheet; matching anchors are lightly highlighted in the document.

ID	Revision note	Anchor
R1	Moved Section 2 toward background technologies and citations; kept system-specific design choices in Section 4.	Background and Related Work
R2	Clarified that RAGAS is limited for ICU Wiki-specific reliability criteria, not simply invalid because model scores are close.	RAGAS
R3	Added explanations for RRF, reranker, and related retrieval terms with references.	Reciprocal rank fusion
R4	Restructured retriever design into problem-oriented subsections.	Retriever design
R5	Moved the Japanese tokenizer example after the process explanation.	Japanese tokenizer
R6	Added a more technical source-authority scoring explanation and a dedicated figure.	Source-authority scoring

ICU Wiki: A Case Study in Reliable Local RAG for University QA

261768 ZHANG Yizhou

Abstract

University information is often fragmented across multiple websites and hidden behind layers of menus, making it difficult for students to find expected answers efficiently. General-purpose LLM chatbots are widely used and good for question answering, but they usually do not have access to institution-internal data that is access-controlled. Retrieval-augmented generation can be a natural approach to address this problem. However, if such a system relies on external API-based models, it may create long-term costs and data-governance concerns.

This thesis presents and evaluates ICU Wiki, a locally hostable RAG-based chatbot for university question answering using locally runnable small language models. The central question is how to define and identify the smallest reliable local configuration, including both the language model and the retrieval-and-orchestration layer around it. We construct a cleaned ICU corpus and evaluate the system on ICU Bench 30, a 30-question benchmark designed around major question categories in the ICU corpus. We test six models under two answer modes and generate 360 answers, which are evaluated using a G-Eval-inspired (Liu et al., 2023) two-judge method across answer acceptance, readability, citation usefulness, and over-answering risk.

The main finding is that average score alone is not sufficient for model selection: smaller models can receive relatively high average scores while still producing unusable or blocking failures under evidence-based QA requirements. Under the proposed reliability criterion, Gemma 4 E4B becomes the smallest observed reliable candidate, while Qwen3.5 9B provides the strongest larger alternative. Overall, this thesis contributes ICU Wiki as a case study of how to build and evaluate an evidence-based local RAG system for university question answering on a fragmented, multilingual institutional corpus, using models small enough to run on local hardware.

1 Introduction

1.1 University QA as an Evidence-Based Problem

University information is often fragmented across multiple systems and hidden behind layers of menus, and ICU is one example of this. We have sites like the e-handbook, the ICU Portal, public web pages, the ICU map, course and syllabus pages, and so on. Students need to go through multiple layers of menus to find the information they need, which is sometimes inefficient. We currently have a keyword-match search function inside the e-handbook, but students often come up with questions in a natural language format rather than just a single keyword. Also, students often need to combine information from several sources instead of just one. However, the current search function inside the handbook is not sufficient for that and requires them to navigate through multiple sources to get the final answer.

LLM-based chatbots have been widely used for question answering and are good at summarizing information from multiple sources. However, general-purpose LLM chatbots usually do not have access to the school's internal data. To address this limitation, RAG is a natural approach for providing extra evidence and generating answers from a custom knowledge base (Lewis et al., 2020). Thus, a RAG-based chatbot can be a practical solution to close this gap.

Students' questions, and the answers they receive, can affect real decision-making, for example, course registration, graduation requirements, senior thesis policy, and so on. A wrong answer could lead students to take wrong actions or make wrong decisions, such as missing a deadline, misusing generative AI tools like ChatGPT, or going to the wrong location when trying to meet with their professor. Therefore, the system cannot only pursue fluency; it also needs to be evidence-based and needs to refuse to answer when the evidence is not sufficient. In this setting, a wrong answer is not only unhelpful, but can also become misleading guidance and thus cause a much more serious result.

Therefore, the key point for the chatbot design is not only whether the system can retrieve relevant information and give a fluent answer, but also whether it can distinguish between sufficient and insufficient evidence and decide whether it should answer the student's question or not. It should only answer when the retrieved institutional evidence supports the answer, and otherwise it should clearly refuse and state that the evidence is insufficient to generate the final answer.

1.2 Local deployment constraints

In terms of deployment, an external API-based service could create excessive long-term costs and potential data-governance risks, especially when the imported data contains internal school information that is meant to be accessible only to ICU students. Therefore, we aim to design a locally deployable system that minimizes long-term API cost while keeping the data inside the school's own environment.

In our case, we want the hardware requirement for deploying the system to not become a major concern. Therefore, we focus on sub-10B small language models.

$$10^{10} \text{ parameters} \times 2 \text{ bytes/parameter} \approx 20 \text{ GB}$$

A 10B-parameter model requires about 20 GB of GPU memory under FP16, which makes deployment under a single general consumer-facing GPU possible, though extra memory for KV cache and runtime overhead might be needed under the real situation. This makes the sub-10B range a practical scope for testing local models under a single-GPU deployment setting.

Smaller models are easier to deploy and may have shorter latency, but there are also important downsides. For example, they may be more likely to over-answer questions and miss key conditions inside the retrieved evidence. They may also be less capable of deciding whether the retrieved evidence is sufficient or insufficient. Their output format can also be less consistent than that of larger models. For a student-facing Q&A use case, these problems are dangerous because the answer may affect real student decisions.

1.3 Research questions

Based on the evidence-safety problem and the answer uncertainty around small, locally hosted models discussed above, this thesis uses ICU Wiki as a case study to examine how reliable a locally

R1

hosted RAG system can be for student-facing university question answering, how reliability should be defined, and how a local configuration that is small but reliable enough can be identified. This thesis uses the following main research question and three subquestions.

Main research question. How can we build and evaluate a local RAG system for student-facing university question answering, and identify a local model configuration that is small but reliable enough for this task?

RQ1. What should count as a reliable answer in student-facing university QA, and how should the question set and benchmark be designed to measure it?

RQ2. How should the retrieval-and-orchestration layer be designed to help small local models answer questions from a multilingual university corpus?

RQ3. Under this evaluation and system design, what local model configuration is small enough while reaching the reliability requirement?

1.4 Contributions

This thesis mainly makes three contributions. First, it presents ICU Wiki as a case-study reference for building a local RAG-based chatbot over a fragmented, multilingual, and multi-format university or institutional corpus. Although the system is built for ICU, a similar setting can be shared by universities and institutions whose information is distributed across multiple domains and stored in various formats.

Second, it proposes a retrieval-and-orchestration layer for this setting, showing how small local language models can be supported by system-side modules and identifying which modules contribute most under small-local-model constraints.

Third, it proposes an evaluation logic for small language models in local RAG-based systems. Instead of fully trusting scores from mainstream RAG evaluation frameworks, it identifies constraints of those evaluation metrics and proposes additional dimensions that need to be evaluated separately or in layers to understand the real reliability and usability of small language models in this setting.

2 Background and Related Work

2.1 Retrieval-Augmented Generation

Retrieval-augmented generation has become a common approach when building an evidence-based chatbot system, as it can provide external evidence by retrieving from an external database with semantic matching and then generate the final answer based on the retrieved evidence (Lewis et al., 2020). In our case, we need evidence-based answer generation in a tightly packed system, so RAG naturally becomes our first option. However, a very simple RAG pipeline itself is not convincing enough, as the retriever could fetch the wrong evidence or the small language model might fail to decide evidence sufficiency, summarize the retrieved passages, or answer the question. Therefore, a carefully designed retriever and orchestration layer, together with careful evaluation, are needed.

R2
R3

2.2 LLM-as-a-Judge

LLM-as-a-judge refers to using a strong language model as an evaluator inside an evaluation framework. For evaluating outputs from a RAG system, answer quality cannot be fully judged by exact keyword matching, because the same question may have multiple acceptable answer forms and language models may not provide the same answer to the same question. With the recent improvement of SOTA-level language models, we believe LLM-as-a-judge provides a relatively consistent and scalable solution to evaluate a large number of generated answers under the same rubric.

2.2.1 RAGAS

Retrieval Augmented Generation Assessment (RAGAS) is an open-source evaluation framework that has been widely used for testing RAG systems (Es et al., 2024). It separates the whole RAG pipeline into parts, including retriever quality, generator quality, and final answer quality. It mainly evaluates the system through four dimensions, including faithfulness, answer relevancy, semantic similarity, and answer correctness. In our case, we started with RAGAS but found that it was not able to adequately evaluate the metrics that are important for our own system and needs, i.e., to distinguish between models and find a model that is small enough while still capable of producing reliable answers.

2.2.2 G-Eval

G-Eval is another LLM-as-a-judge evaluation framework that was built not only for evaluating RAG systems, but was first designed to mainly test a model's summarization and dialogue-generation ability (Liu et al., 2023). G-Eval's logic is simpler with less calculation. Instead, it lets us define our own evaluation criteria and scoring logic, send them to a strong LLM, and ask it to score the output using a structured form and a numerical score range.

2.3 Small Language Models

Small Language Models (SLMs) refer to language models that are much smaller than frontier LLMs, such as GPT-5-class models or Claude Opus-class models. There is no strict size boundary, but in our case, we use SLMs smaller than 10B as our testing scope. SLMs are normally cheaper, faster, and easier to self-host because they contain far fewer parameters inside the model. However, they are also weaker in reasoning ability, consistency, and generation stability because of their smaller parameter count.

2.4 Retrieval Ranking and Authority Scoring

2.4.1 Reciprocal Rank Fusion (RRF)

Reciprocal Rank Fusion (RRF) is a method for merging and combining the results from multiple retrievers or ranking systems into one final ranking (Cormack et al., 2009). Suppose each retriever returns its own ranked list of document chunks. RRF calculates the score as:

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{k + \mathrm{rank}_r(d)}$$

where $\mathrm{rank}_r(d)$ is the rank of document chunk d in retriever or ranking system r, and k is a constant. For each document chunk, it calculates the sum of the scores from each retriever and gives a final ranking as a list. In this case, document chunks that appear in multiple retrievers, and chunks that appear near the top, will get a higher final score.

2.4.2 Reranker

A reranker is a second-stage model that takes the candidate document chunks retrieved by the first-stage retriever and reorders them more carefully (Nogueira and Cho, 2019). It is different from first-stage retrievers like BM25 or dense vector search, which prioritize computational efficiency because they need to deal with a large amount of text. The reranker only works on candidates that have already been pre-selected by the first-stage retriever, and it is applied to these candidates to calculate a score that estimates their relevance to the query in a more precise way and produce a refined ranking. Nowadays, rerankers made for retrieval-augmented generation are often transformer-based models. They take the query and the candidate chunk as input and output a relevance score.

2.4.3 Authority and source-aware scoring

2.5 Agentic Frameworks

3 Task and University Corpus

3.1 ICU university corpus

As mentioned above, the ICU corpus is fragmented, multilingual, and multi-format. It contains ICU-related institutional knowledge spread across multiple domains and represented in different formats. We mainly prepare the data from five source groups: course records, ICU e-handbook pages, public web pages under ICU-related domains, portal-derived items, and PDF documents discovered from those pages or links. We treat PDF documents as a separate source group because many policy or guide materials are not written directly inside HTML pages but are attached as downloadable files.

Our goal is to include as much data from ICU-related official or linked sources as possible, so that the system can cover a wider range of student questions.

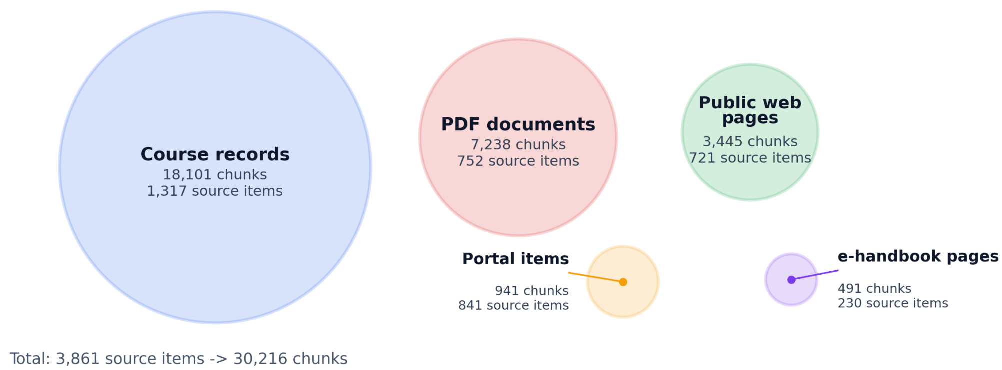

Figure 1: ICU corpus composition by generated chunk count.

Source group	What it contains	Example text
Course records	Course number, instructor, term, credits, syllabus sections, grading policy	"Course No: ELA010; Registration No: 10101; Term: Spring 2026; Instructor: STAFF; Credits: 3."
e-handbook pages	Academic rules, graduation requirements, registration, thesis policy	"The graduation requirements differ depending on the year of admission."
Public web pages	Public office pages, student support, financial information, service pages	"Payment of University Fees; Paying Tuition and Dormitory Fee; FAQ Tuition."
Portal items	Portal notices, course registration announcements, deadlines, calendar events	"Online input December 3 (Wed) 9:00 – December 4 (Thu) 14:30."
PDF documents	Attached guides, forms, dormitory rules, policy PDFs, newsletters	"Please apply to bring in items after entering the dormitory."

Table 1: Examples of source items in the ICU corpus.

3.2 Corpus construction

We collect raw data from the sources mentioned above. However, raw web data contains many headers, boilerplate sections, navigation texts, and repetitive text blocks that appear across web pages because of the HTML structure. Therefore, we normalize the collected data by keeping useful metadata and removing unnecessary text.

We then apply source-aware chunking. Instead of using one fixed chunking strategy for every source, we use different chunking strategies for different source types. For each chunk, we keep the citation and metadata, so that when the model references a chunk, the original source link can still be shown to users. Finally, we input all chunks into the embedding and retrieval-indexing pipeline.

3.3 ICU Bench 30

R4

R6

ICU Bench 30 is a benchmark question set that contains 30 questions designed to evaluate student-facing university QA over the ICU corpus. We designed this question set so that it covers major categories inside the ICU corpus data and student question types. The major question categories include course (5), registration (4), policy (4), and curriculum (4). The question set covers three languages: English (20), Japanese (7), and Chinese (3), based on the multilingual student composition at ICU. The goal is not only to mirror the real-life needs of ICU students, but also to test whether the model can handle different complex question situations. We also included non-answerable questions (Q13, Q15, Q25, Q26, and Q27) on purpose to test whether the model can answer from evidence, use good citations, and avoid over-answering.

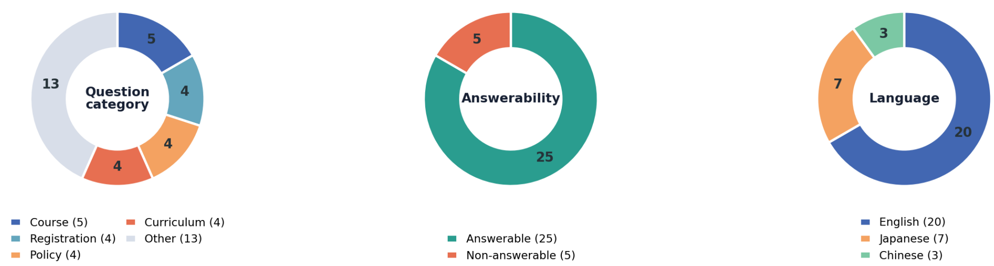

Figure 2: ICU Bench 30 composition by question category, answerability, and language.

4 System Design

4.1 System overview

ICU Wiki is built as a local RAG system with three main parts: a cleaned ICU corpus, a retriever that selects top evidence chunks, and an answer generator based on a local language model. The retriever went through four major updates after the basic hybrid retrieval baseline: Japanese FTS, metadata/entity matching, reranking, and source-authority scoring. The following sections describe the retriever design and the two answer orchestration modes used in the evaluation.

4.2 Retriever design

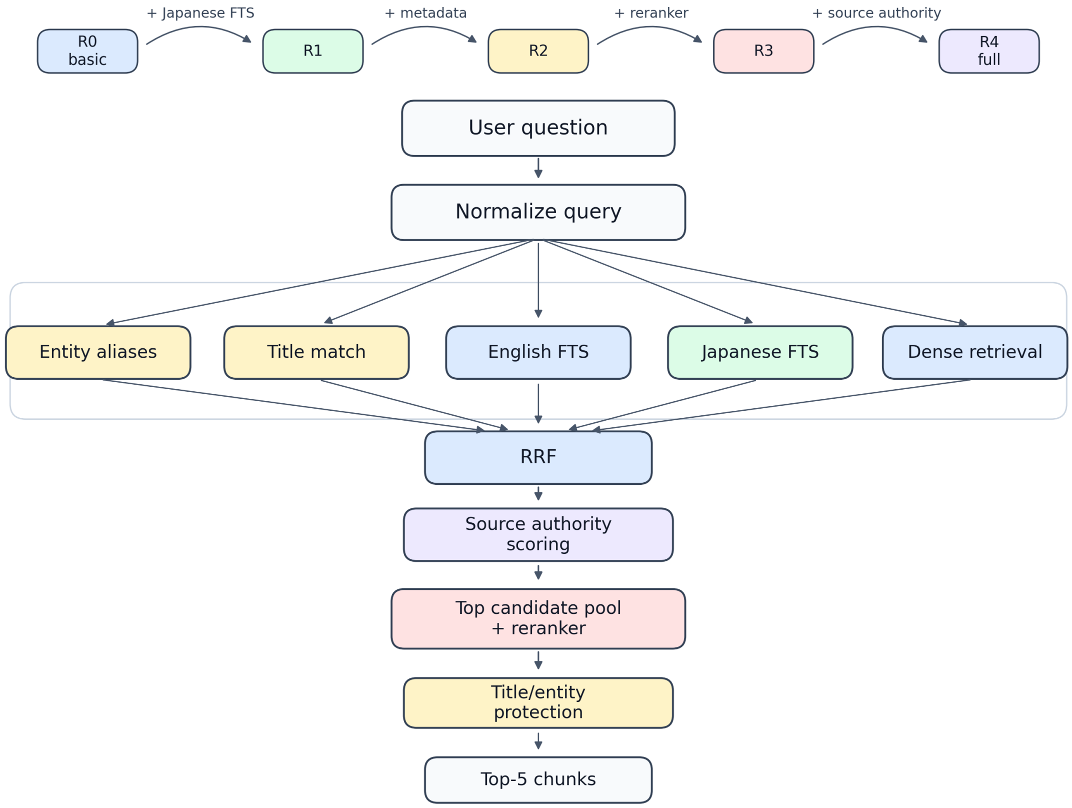

Figure 3: Retriever iteration.

4.2.1 Basic hybrid retrieval

R5

Implementing the retriever on the ICU corpus was challenging because the corpus contains multiple categories, document types, and two languages. First, we tried the simplest hybrid retrieval method, which combines keyword retrieval and dense retrieval based on the BGE-M3 embedding model, the same model we used to embed all the chunks in the ICU corpus. This worked for some basic natural-language search, but it was unreliable for questions in Japanese and for questions that contained specific terminology related to ICU or technical terms, including professors' names, ICU-specific terminology, and numbers.

4.2.2 Japanese tokenizer

We analyzed the bad cases and found that the simple text-search configuration from PostgreSQL worked well for English sentences because English words are separated by spaces. However, for Japanese sentences, it did not know how to separate the sentence into proper words. Therefore, for keyword matching, especially when a query contained a very specific keyword that gave a strong hint about which corpus area it might relate to, the result was poor. One example was 鏑木先生が担当している授業は何ですか？ ("Which classes does Professor Kaburagi teach?"). The English or romanized version could sometimes work better, but the Japanese version was much less reliable before the Japanese retrieval improvements.

We therefore upgraded the lexical side with PostgreSQL FTS and an additional Janome Japanese tokenizer, so that the system could split Japanese question sentences into more proper word units and produce better keyword-matching results.

Before using the Japanese tokenizer, the system could only process the sentence as a whole or use weak CJK lexical separators:

> **As a whole sentence:**
> 鏑木先生が担当している授業は何ですか？
>
> **With CJK separators:**
> 鏑木 / 木先 / 先生 / 生が / が担 / 担当 / 当し / して / てい / いる / る授 / 授業
> (This lists many character-combination possibilities, which is not very efficient.)
>
> **With Janome tokenizer:**
> 鏑木 / 先生 / 担当 / 授業

4.2.3 Entity aliases and title protection

For certain terminology matching, because specific terms can provide a strong hint during retrieval, we also built an alias table from course and person metadata. This alias table improves exact matching for data chunks linked to certain entities, especially for teacher and course-name-related lookups. The goal was to keep chunks with specific titles, numbers, names, course names, or professor names from being buried under chunks that matched more semantically but were not the desired chunks.

4.2.4 Reranker

We then expanded the candidate pool and added a reranker. This allowed the system to first collect a broader set of candidate chunks and then rerank them by comparing the question and candidate passages more directly.

4.2.5 Source-authority scoring

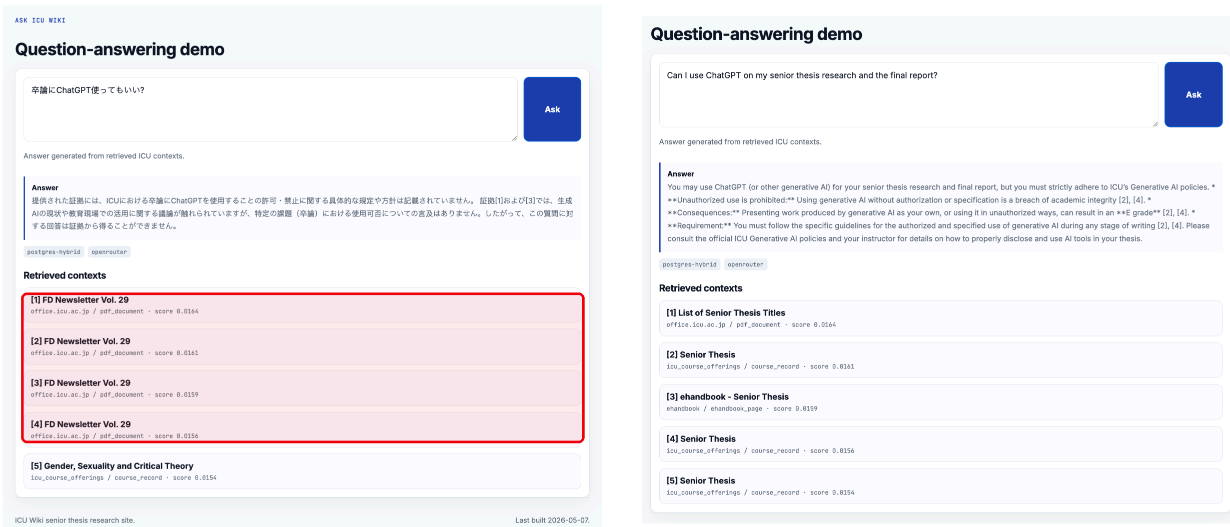

Figure 4: Qualitative comparison from the earlier runtime: Japanese and English ChatGPT thesis-policy queries retrieved different evidence under the same retriever.

After that, we found another type of bad case. For questions such as 卒論に ChatGPT を使っても大丈夫ですか？ (“Is it okay to use ChatGPT for a senior thesis?”), the retriever tended to retrieve chunks from newsletters or event-like pages. Those chunks surely contained related words and related content, but for questions related to policies, we wanted evidence from more authoritative policy-related materials, such as the e-handbook.

Therefore, we added source-authority scoring. Instead of only asking whether a chunk is relevant to the query, this scoring also considers the source type and the broad query category. For example, policy and procedure questions should prefer e-handbook, official web pages, or portal notices, while course-specific questions should prefer course records. This shifted the retriever from finding merely relevant chunks toward selecting evidence that is more appropriate for a specific student-facing question.

In the backend retriever, we calculate an additional score between the question and the retrieved chunks before the reranker. Each chunk already has a source type assigned during corpus construction. For each question, several flags are triggered when certain keywords are detected. If those flags match with a certain source type, the score is increased; otherwise, the score can be decreased or remain the same. In the current runtime, the reranker then reranks the top 40 candidates.

Figure 4 shows a bad case from an earlier runtime. Here we can see that Japanese and English queries can retrieve different evidence under the same retriever, and policy questions need more source authority rather than only keyword relevance. Although the meaning of the Japanese and

English questions is almost the same, the Japanese request only retrieved newsletter-like content, while the English request retrieved sources from ICU official web pages and the e-handbook.

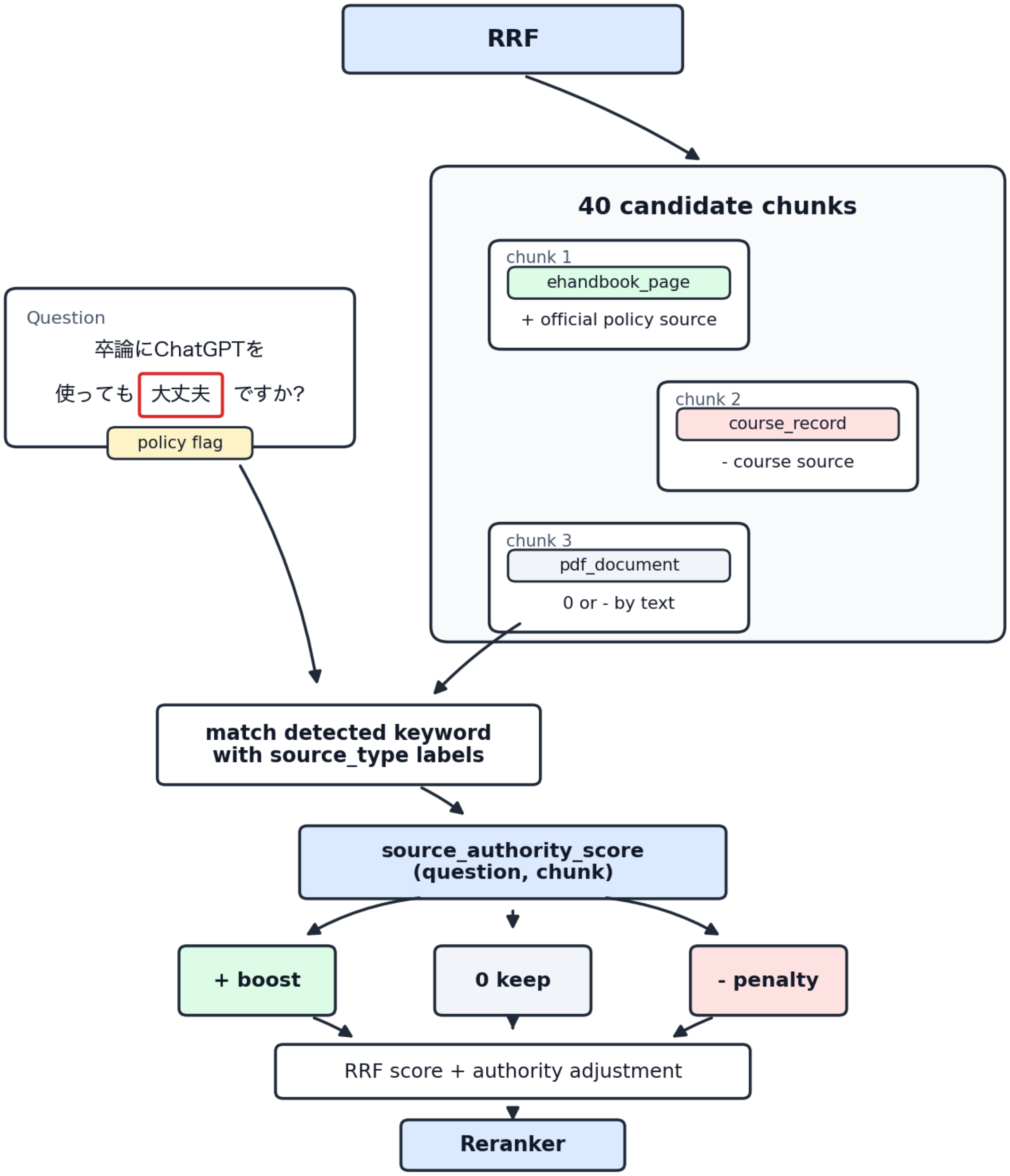

Figure 5: Source-authority scoring inside the backend retriever.

4.3 Answer orchestration modes

4.3.1 Single-shot mode

In our system design, we use single-shot mode as the baseline. Under single-shot mode, each question passes through one fixed retriever and one fixed retrieval action, and the top-five evidence chunks are sent directly to the answer generator. As a result, it is easier to compare the difference between each model's pure generation performance, because the retriever side is fixed and will not change because of an agent loop.

4.3.2 Agentic Retrieval Mode

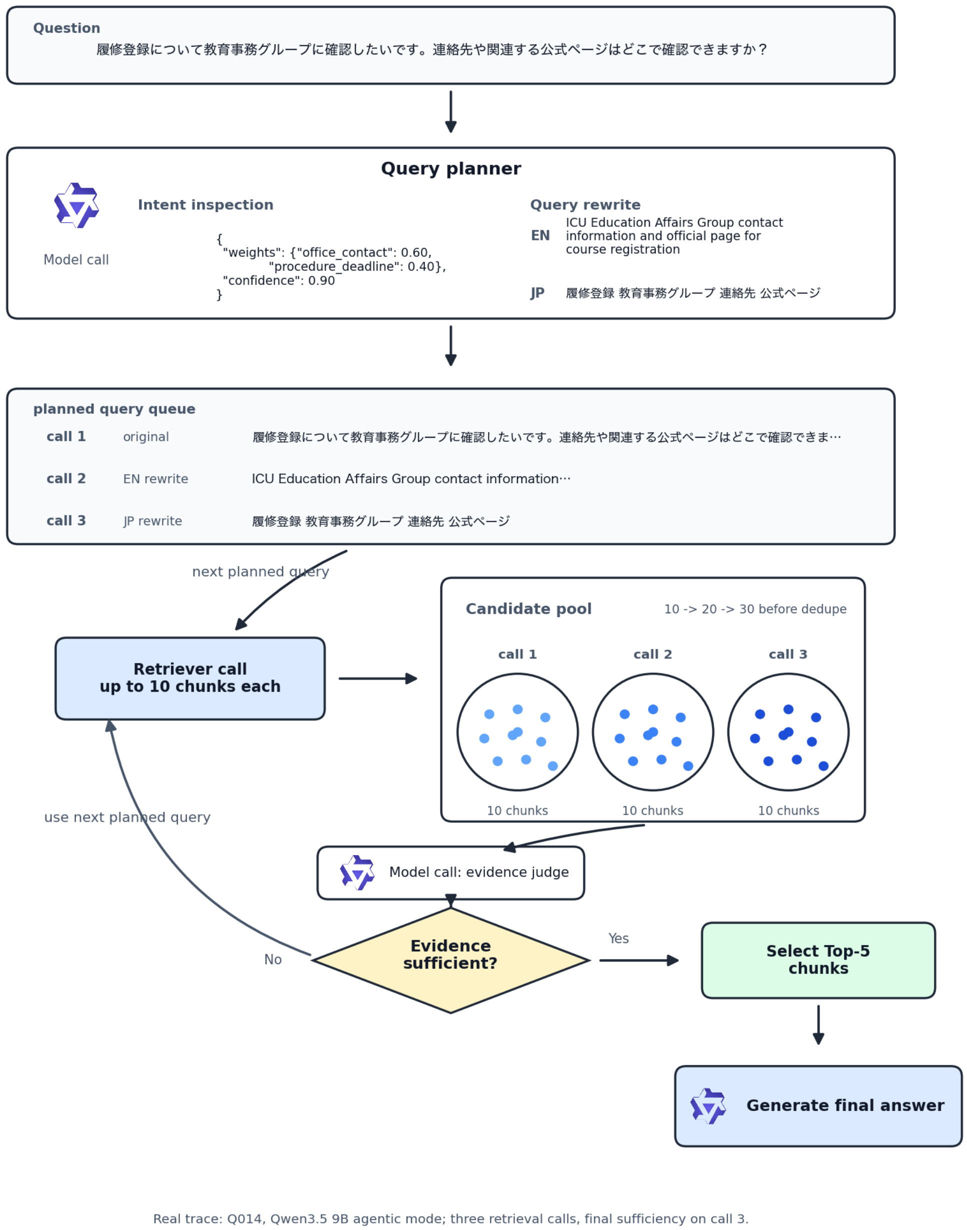

Figure 6: Agentic retrieval framework using a real ICU Bench 30 trace.

Considering recent progress in agentic retrieval methods, where the model can rewrite the query if the query is bad and perform multiple rounds of search if it thinks the evidence is not enough, we propose adding an agentic mode to the current baseline. Agentic mode can rewrite the input question into both English and Japanese variants, and it can perform multiple rounds of search if it thinks the retrieved evidence is not enough to generate the final answer. We limit the maximum number of search rounds to three to keep the system delay within a controllable scope.

However, agentic retrieval relies heavily on the model's own performance. If the controller model behind it is not strong enough, it may misjudge evidence sufficiency. Under multiple rounds of retrieval, it may also bring in more irrelevant chunks, which can lead to a worse answer. Furthermore, agentic mode can significantly increase latency. Therefore, we cannot assume that agentic mode is always better, and careful comparison needs to be performed between models and generated answers under different questions. We then decide which mode is more suitable for our system design.

5 Evaluation Methodology

5.1 What counts as a reliable answer?

Under the university QA situation, defining what should count as a reliable answer is important. We propose four criteria that both the model and the retriever need to meet:

- **Evidence-based.** The generated answer must be based on the retrieved evidence instead of being filled by the model's own internal knowledge or unrelated evidence. This is extremely important, especially for questions related to school policy, registration deadlines, and course requirements.
- **Summarize instead of copied.** The answer needs to be rewritten in the model's own words and understanding, and it needs to stay within a reasonable length constraint. It should not give a short answer and then directly copy and paste all five contexts into its generated answer, which we have observed in the 0.8B model's generated results. The model needs to have a certain level of summarization ability to make the answer easy for students to read. Otherwise, there is no meaning in having the generator instead of only using the retriever itself.
- **Citation useful.** The generated answer needs to include useful citations. It needs to have citation marks, and each citation mark needs to be attached to the correct piece of generated text, so that students can check the original source while reading the generated answer.
- **No blocking case.** We also want to avoid blocking cases, such as endless repetition loops from the model, truncation, gibberish, extreme hallucination, or answers based on insufficient evidence. If a model shows these blocking cases, even if it has a relatively high average score, we consider it not usable because it is not reliable enough.

5.2 Limits of standard RAG evaluation

At first, we used RAGAS as a mainstream evaluation method to evaluate the whole-system RAG performance. However, after running it across our six models, we found that the score differences were very small, making it difficult to determine the real model differences or set a threshold for deciding which model could be used for the actual system generation.

Furthermore, under these very close RAGAS scores, we found that the actual generated answers differed more than the scores suggested. For example, the 0.8B model's readability and summarization ability were low. Even when it produced an answer that was correct enough and faithful to the retrieved context, it could still be unusable from a human perspective because it was not readable and was inconsistent in format. Therefore, we found that some metrics are not reflected by the RAGAS evaluation framework. The current RAGAS evaluation framework has certain constraints, especially for readability, model generation consistency, and blocking-case detection.

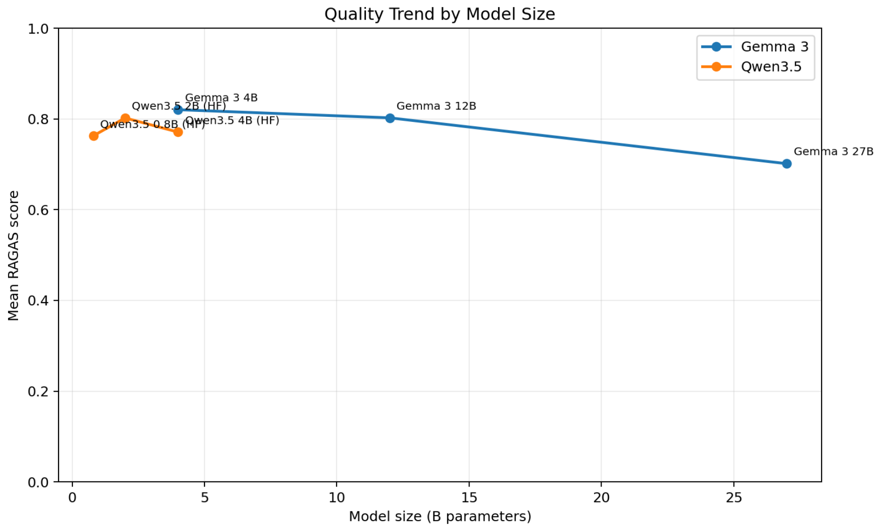

Figure 7: Early mean scores from RAGAS by model size, a diagnostic reference for the limits of RAGAS scoring.

Language following is another usability issue that is not directly captured by a single average RAG score. As a post-hoc diagnostic, we checked whether each generated answer used the same main language as the input question.

5.3 G-Eval-inspired answer-quality evaluation

We then fed these answer pairs to GPT Pro and DeepSeek V4 Pro to let them blindly evaluate the answers. The judges gave scores from 0 to 2 on four dimensions: answer acceptance, readability, citation usefulness, and over-answering risk. To reduce possible evaluator bias from one model family, we intentionally chose two capable models from two different model families. The result shows that the stronger local models generally receive higher human-facing quality scores, but the difference between single-shot and agentic mode is not the same across model sizes.

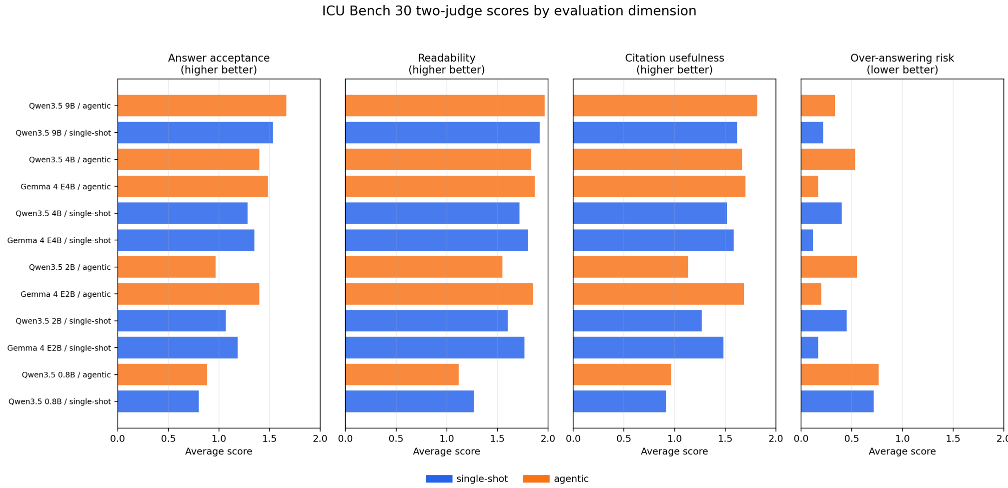

Figure 8: ICU Bench 30 average scores under two judges across four evaluation dimensions.

5.4 Working reliability requirement

As we evaluate model output by assigning scores from 0 to 2 on each evaluation dimension and then normalize the final quality score, we use 0.8 as a practical bottom line to filter out models that are obviously not capable enough for this task. Around this level, many models can still meet a 0.7 threshold, but their real outputs are not stable enough. This is especially clear for the 0.8B and 2B models, where we observed poor readability, direct copying from the retrieved context, and language-following failures.

Meanwhile, 0.9 seems to be an overly conservative choice, as it will mostly allow only the large models or strongest models to pass. However, meeting the 0.8 threshold is still not enough to confirm the model's usability. As mentioned above, blocking cases can severely damage user experience and create potential risk. Therefore, in our definition, a workable model condition needs to meet the 0.8 quality threshold and also have zero blocking cases.

5.5 Blocking cases

In this retriever ablation, we define a blocking case as an answer where `answer_acceptance` is `0` from any of the two judges, `human_quality_score` is below `0.50`, or the judge notes/tags indicate repetitive or truncated output.

6 Experiments and Results

6.1 Model and mode comparison

For the model comparison experiment, we provided each ICU Bench 30 question and its retrieved chunks to six models, including the Qwen3.5 family from 0.8B to 9B and the Gemma 4 2B and 4B models. Each model generated answers in two modes: single-shot and agentic. As a result, we received 360 answer pairs.

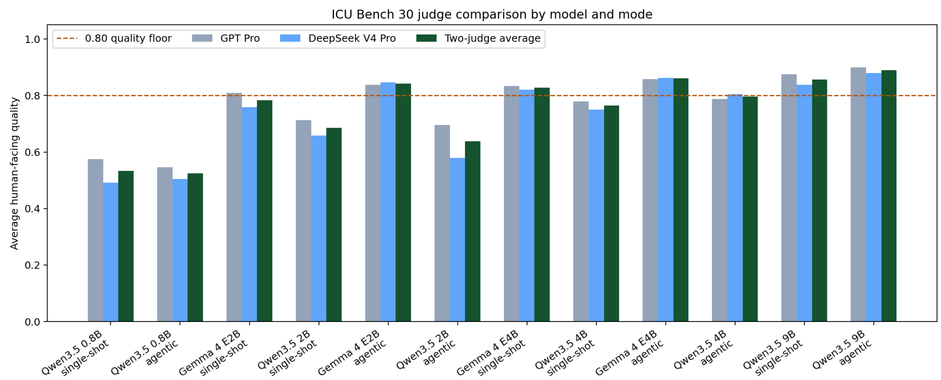

Figure 9: ICU Bench 30 judge comparison by model and answer mode.

We compared the quality scores produced by two SOTA-level judge models, GPT Pro and DeepSeek V4 Pro, and took their average to reduce single-judge model bias. Across the 12 model/mode conditions, GPT Pro gave a higher average score in nine conditions, while DeepSeek V4 Pro gave a higher average score in three conditions. This variability is the reason we use the average of the two judges instead of relying on a single model family.

If we set the working threshold to 0.8 on the two-judge average score, the conditions that reach this threshold are Gemma 4 E2B in agentic mode, Gemma 4 E4B in both single-shot and agentic modes, and Qwen3.5 9B in both single-shot and agentic modes.

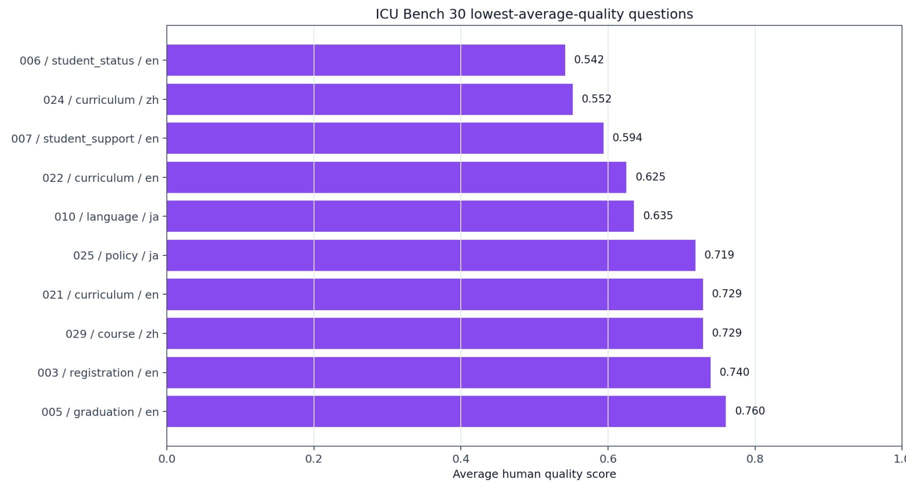

Figure 10: ICU Bench 30 lowest-average-quality questions.

6.2 Smallest reliable configuration

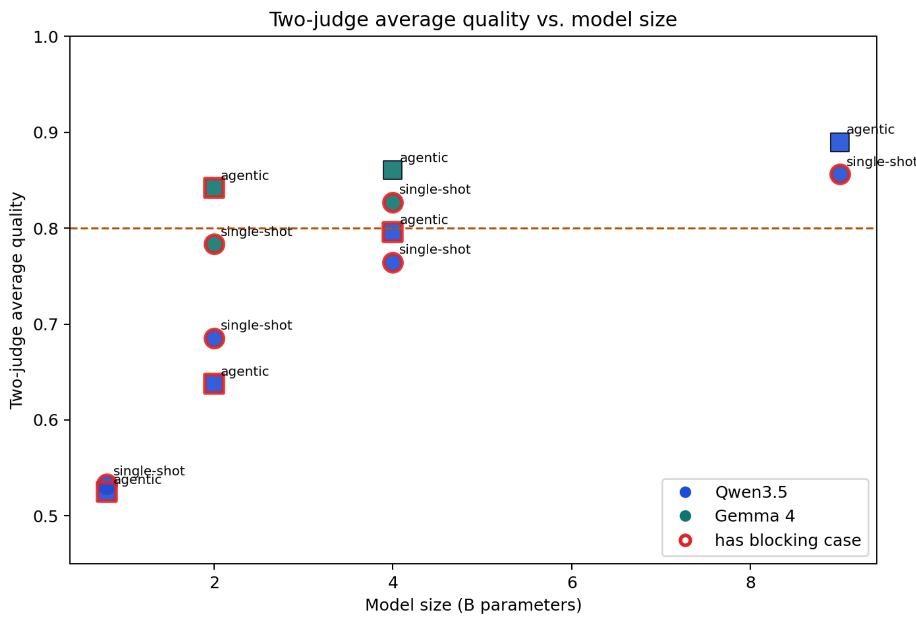

Figure 11: Average quality with blocking cases marked under two judges across model sizes and answer modes for the working-threshold screen.

The previous score-only comparison is not enough to clarify the practical difference between the models. Because blocking cases matter for an evidence-based chatbot, we add blocking-case analysis on top of the average score. After marking blocking cases across these model/mode conditions, we find that Gemma 4 E4B agentic mode and Qwen3.5 9B agentic mode are the only two conditions that are above the quality threshold while also having zero blocking cases.

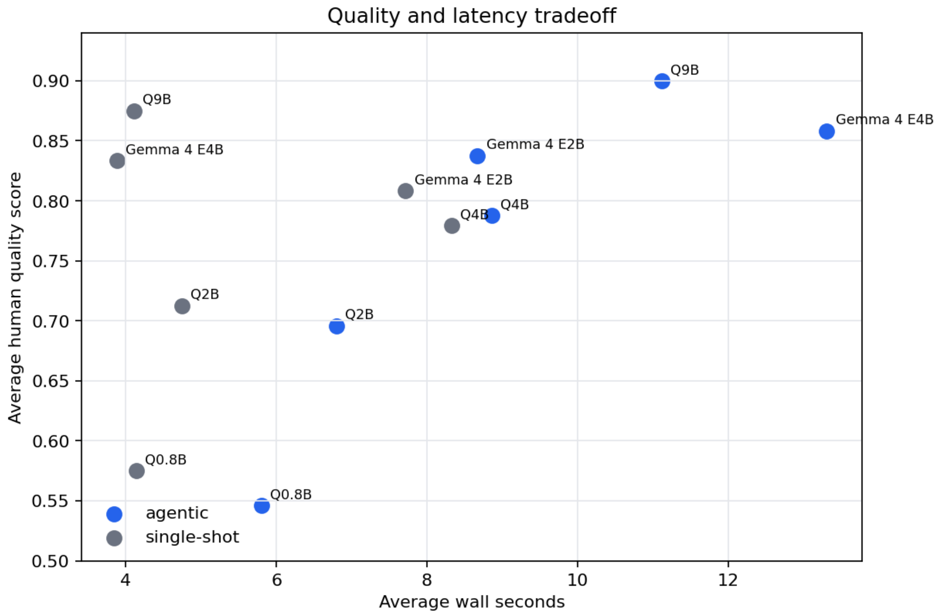

Figure 12: ICU Bench 30 quality-latency tradeoff by model and answer mode.

With Gemma 4 E4B and Qwen3.5 9B under agentic mode passing our reliability requirement, we then consider latency and hardware constraints. Qwen3.5 9B provides the strongest deployment choice if the hardware is strong enough and has enough GPU memory. Gemma 4 E4B, on the other hand, can fit into a tighter hardware constraint, although it showed higher latency in this run.

6.3 Retriever ablation

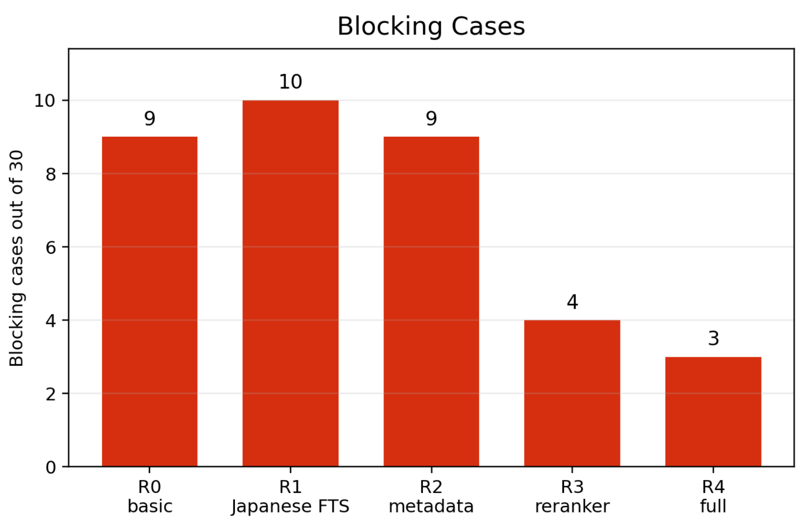

Figure 13: Blocking cases across retriever versions.

6.3.1 R0 to R1 (added Japanese FTS)

Blocking cases decreased

- `s30_023`: ICU に入学した後、必ず履修しなければならない科目は何ですか？どの公式ページを見るべきですか？
- `s30_029`: 鏑木先生教哪些课？请只根据检索到的 ICU 课程资料回答。

Blocking cases increased

- `s30_018`: MIURA 先生って、どの日本語の授業を担当していますか？レベルも少し知りたいです。
- `s30_025`: 大学の卒論にチャット GPT を使っても大丈夫ですか？
- `s30_027`: 我可以在 ICU 的毕业论文里使用 ChatGPT 吗？如果资料不够，请明确告诉我。

6.3.2 R1 to R2 (added metadata/entity matching)

Blocking cases decreased

- `s30_018`: MIURA 先生って、どの日本語の授業を担当していますか？レベルも少し知りたいです。
- `s30_028`: 鏑木先生が担当している授業は何ですか？

Blocking cases increased

- `s30_007`: I feel overwhelmed and want to talk to someone, but I am worried about privacy. Where can I find official information about counseling or student support at ICU?

6.3.3 R2 to R3 (added reranker)

Blocking cases decreased

- `s30_007`: I feel overwhelmed and want to talk to someone, but I am worried about privacy. Where can I find official information about counseling or student support at ICU?
- `s30_009`: I am graduating from undergraduate school but continuing to ICU graduate school. What should I know about my IT account or services after graduation?
- `s30_025`: 大学の卒論にチャット GPT を使っても大丈夫ですか？
- `s30_027`: 我可以在 ICU 的毕业论文里使用 ChatGPT 吗？如果资料不够，请明确告诉我。
- `s30_030`: After I select courses on the registration website, how can I check whether my registration is actually complete?

Blocking cases increased

- None.

6.3.4 R3 to R4 (added source authority)

Blocking cases decreased

- `s30_014`: 履修登録について教育事務グループに確認したいです。連絡先や関連する公式ページはどこで確認できますか？

Blocking cases increased

- None.

6.4 Effect of answer orchestration mode

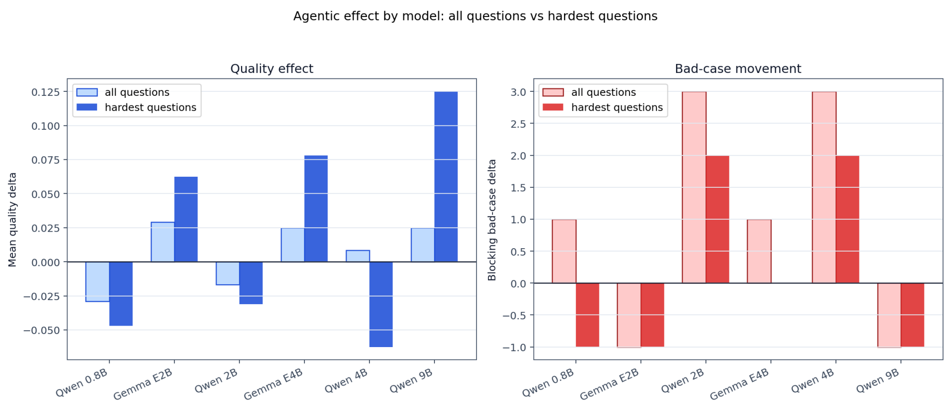

Figure 14: Agentic effects by model on blocking-case changes under ICU Bench 30.

We want to compare whether agentic mode improves answer quality generally across models and whether it changes the number of blocking cases. However, the result does not show a universal pattern that agentic mode is always better. Among these models, Gemma 4 E2B, Gemma 4 E4B, and Qwen3.5 9B show quality improvement in agentic mode, but Qwen3.5 0.8B and Qwen3.5 2B become worse. On the blocking-case side, Qwen3.5 2B and Qwen3.5 4B show more blocking cases in agentic mode, while Gemma 4 E2B and Qwen3.5 9B show fewer blocking cases. This suggests that agentic retrieval may help stronger or more stable models, but it is not automatically helpful for every small model.

6.5 Language-following diagnostic

Language following means that the input language and the output language should be aligned. In other words, if the user asks a question in Japanese, the answer should also be in Japanese; if the user asks in Chinese, the answer should also be in Chinese.

Language following is another important part of model evaluation, because an output in a language that the user does not use may lead to confusion. We evaluated the model's language-following capability on ICU Bench 30, which contains English, Japanese, and Chinese questions. Among these three language groups, Chinese questions appear to be the hardest case. Only the 9B model consistently followed the input language for Chinese questions, while smaller models often changed the answer language.

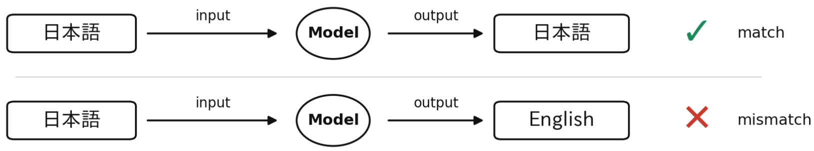

Figure 15: Concept of the language-following diagnostic.

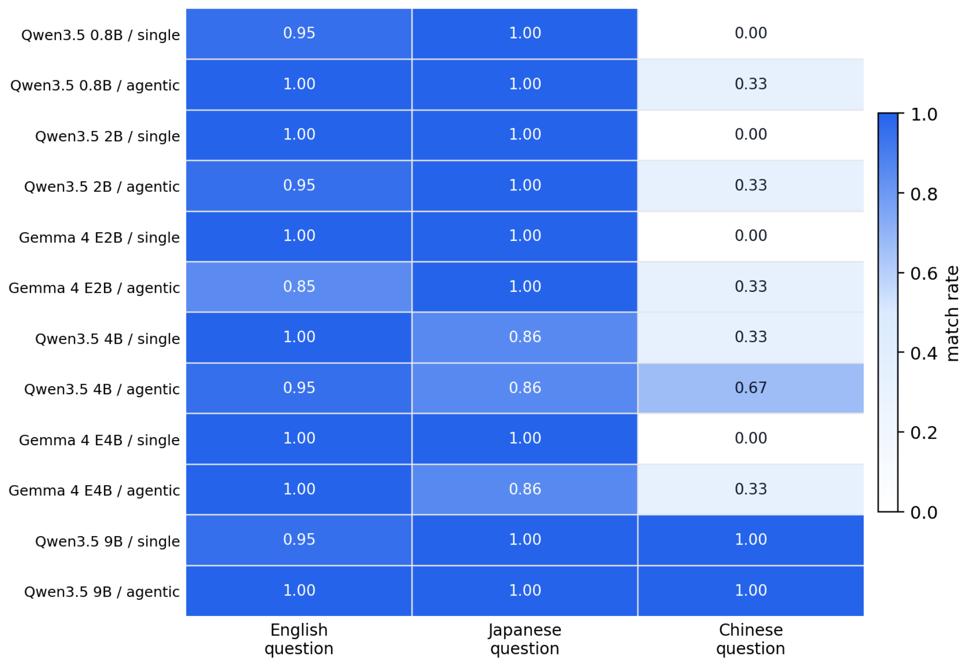

Figure 16: Input and output language match rate by question language, model, and answer mode.

7 Bad Case Analysis and Discussion

7.1 Observed failure patterns in small models

From the observed small-model outputs, the main failure patterns are uncontrolled self-repetition near the end of the answer, direct copying from retrieved context, and difficulty summarizing across multiple easily confused sources.

7.1.1 Uncontrolled self-repetition near the end of the answer

Case. 023.

Answer from Qwen3.5 0.8B, agentic mode.

ICU の入学後、必ず履修しなければならない必修科目は「リベラルアーツ英語プログラム（ELA）」です。[1] Title: ICU の入試について｜国際基督教大学（ICU） Source: https://www.icu.ac.jp/admissions/undergraduate/exam Evidence: 一般選抜、総合型選抜 <4 月入学 >、総合型選抜 <4 月/9 月入学 >、EJU（日本留

学試験）利用、学校推薦型選抜 <4 月入学専願 > を受ける場合、卒業要件上の必修語学科目は「リベラルアーツ英語プログラム」（ELA: English for Liberal Arts Program）となります。[2] Title: ICU の入学について｜国際基督教大学（ICU）Source: https://www.icu.ac.jp/admissions/undergraduate/undergraduate Evidence: 入学者選抜に関するお問い合わせ先（一般選抜、転編入学制度、総合型選抜 <4 月入学 >、総合型選抜 <4 月/9 月入学 >、EJU（日本留学試験）利用、学校推薦型選抜 <4 月入学専願 >、研究生、聴講生・科目等履修生）は、アドミッションズ・センターへお問い合わせください。[3] Title: ICU の入学について｜国際基督教大学（ICU）Source: https://www.icu.ac.jp/admissions/undergraduate/undergraduate Evidence: 一般選抜、総合型選抜 <4 月入学 >、総合型選抜 <4 月/9 月入学 >、EJU（日本留学試験）利用、学校推薦型選抜 <4 月入学専願 > を受ける場合、卒業要件上の必修語学科目は「リベラルアーツ英語プログラム」（ELA: English for Liberal Arts Program）となります。[4] Title: ICU の入学について｜国際基督教大学（ICU）Source: https://www.icu.ac.jp/admissions/undergraduate/undergraduate Evidence: 一般選抜、総合型選抜 <4 月入学 >、総合型選抜 <4 月/9 月入学 >、EJU（日本留学試験）利用、学校推薦型選抜 <4 月入学専願 > を受ける場合、卒業要件上の必修語学科目は「リベラルアーツ英語プログラム」（ELA: English for Liberal Arts Program）となります。[5] Title: ICU の入試について｜国際基督教大学（ICU）Source: https://www.icu.ac.jp/admissions/undergraduate/exam Evidence: 一般選抜、総合型選抜 <4 月入学 >、総合型選抜 <4 月/9 月入学 >、EJU（日本留学試験）利用、学校推薦型選抜 <4 月入学専願 > を受ける場合、卒業要件上の必修語学科目は「リベラルアーツ英語プログラム」（ELA: English for Liberal Arts Program）となります。[6] Title: ICU の入学について｜国際基督教大学（ICU）Source: https://www.icu.ac.jp/admissions/undergraduate

Answer from GPT Pro (reference).

検索資料で確認できる必修要件は、保健体育の「健康科学（Health Science）」1 科目 1 単位と「PE エクササイズ I・II・III」3 種目合計 1 単位、および入試区分等により決まる必修語学（ELA または JLP）です。一方で、入学後に必ず履修する全科目の完全な一覧はこの証拠だけでは断定できません。公式ページは「卒業要件」「保健体育」「ICU の入試について／日英バイリンガル教育」や在学生向け eHANDBOOK を確認してください。

In this case, the model kept doing self-repetition over a cited piece of text until it ran out of the maximum output token budget.

7.1.2 Direct copying from retrieved context

Case. `014`.

Answer from Qwen3.5 0.8B, single-shot mode.

履修登録に関する公式確認方法と連絡先は、以下の情報から確認できます。[1] 教務グループからの重要なお知らせ（春学期登録版）[1] 教務グループからの重要なお知らせ（春学期登録版）* 履修証明書の申請は、卒業年度最終学期登録日から登録変更期間終了日までの間に申請する必要があります。* 詳細は 4 月 5 日（金）に ICU Portal お知らせを出していますので、確認してください。* 問い合わせ先：教務グループ ea-group@icu. ac. jp [2] 教務グループからの重要なお知らせ（春学期登録版）[2] 教務グループからの重要なお知らせ（春学期登録版）* 履修証明書の申請は、卒業年度最終学期登録日から登録変更期間終了日までの間に申請する必要があります。* 詳細は ICU Portal お知らせを出していますので、確認してください。jp [3] 履修登録 [3] 履修登録 * 履修登録の手続きは、Registration Website でその学期に履修する科目を入力します。* 登録または登録変更を行った後は、必ず Registration Website の「Registration Results」で登録結果を確認してください。* 履修登録および登録の確認は学生本人の責任であり、怠ったために生じた問題について大学は一切責任を負いません。* 履修登録の手順については、(6) 参考リンクの各項目をよく読むこと。ただし一部の科目は、(b) 申請書による履修登録の説明に則って登録を申請してください。* 提出期間は登録日〜登録変更期間最終日 10:00 です。[4] 履修登録 [4] 履修登録 * 履修登録日は 12 月 4 日（水）と 12 月 5 日（木）の 2 日間です。詳細は、追って ICU Portal に掲載する「

Answer from GPT Pro (reference).

履修登録・成績・卒業要件に関する問い合わせ先として、教務グループ（Educational Affairs Group）ea-group@icu.ac.jp が示されています。関連する公式ページは、eHANDBOOK の「履修登録」ページと、各学期の ICU Portal/eHANDBOOK に出る教務グループからの科目登録に関する重要なお知らせです。履修登録ページでは、Registration Website での入力、アドヴァイザー面談・オンライン承認、Registration Results の確認が説明されています。

In this case, the model took a shortcut and gave a short answer structure while copying and pasting retrieved context into its output. This can lead to a relatively high score under RAGAS-style evaluation because the output contains correct evidence, but this answer format is not preferred because its readability is poor.

7.1.3 Difficulty summarizing across multiple easily confused sources

Case. `017`.

Answer from Qwen3.5 0.8B, single-shot mode.

Based on the provided evidence, the grade evaluation for **Statistical Information Literacy** (Course S2: GES039) is determined by the following components: * **Exam (20%):** The syllabus explicitly states that the exam consists of **60% of the total grade** (100% - 40% - 20% - 20%). * **Report (40%):** The syllabus requires students to submit a **Report** (40% of the total grade). * **Daily quiz (20%):** Students must participate in **daily quizzes** (20% of the total grade). * **Class participation (20%):** Students must attend and participate in **classes** (20% of the total grade). **Conclusion:** The grade is calculated based on the sum of the exam, report, daily quiz, and class participation. The syllabus does not specify a single "required work" other than the report and daily quizzes; it requires students to complete the daily quizzes and submit the report as part of the assessment.

Answer from GPT Pro (reference).

> The grading depends on the section. In Spring 2026, the Japanese section is evaluated by Exam 20%, Report 40%, Daily quiz 20%, and Class participation 20%; in Autumn 2026, the English section is evaluated by Active participation 15%, Application workshops 35%, and an end-of-term data analysis project 50%. The syllabus descriptions suggest work in basic statistics, sampling/data processing, probability, distributions, sample data analysis, and computer/statistical-software-based data analysis.

In this case, the same class name and course number appeared across multiple course records or terms, and the grading methods differed between them. The 0.8B model failed to distinguish these records clearly and mixed the information together in its output.

7.2 Why average scores hide deployment risk

Small models' average scores can get close to larger models' scores. However, this does not necessarily suggest that they are capable enough to be deployed in a real situation. Average scores capture average-case usefulness, while answer consistency and blocking cases strongly affect user experience and can cause potential risk by providing unusable answers. Therefore, in our consideration, the smallest high-scoring model is not necessarily the smallest reliable model.

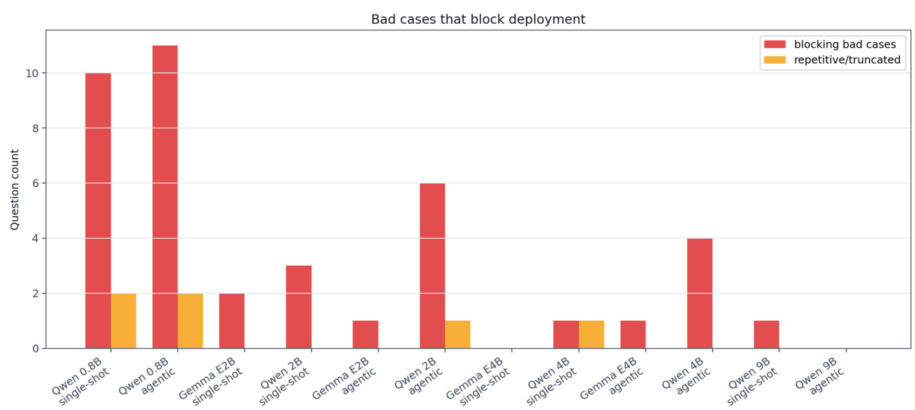

Figure 17: ICU Bench 30 blocking cases by model and answer mode.

7.3 Feasibility boundary for local university QA

In certain narrow cases for university QA, small models have the potential to be workable when there is a strong retrieval-and-orchestration layer outside the model. However, the real boundary is not whether the small model can answer most questions, but whether it can avoid certain blocking cases. Therefore, average quality may saturate earlier than reliability.

7.4 Output-control risk in retrieval-and-orchestration layers

Our live testing outside the benchmark question set also triggered a format-control failure while in agentic mode. In this example, in-between process-like text appeared inside the final answer, which led it to a blocking case. It shows a potential risk of the retrieval-and-orchestration layer: while in agentic mode, it could possibly introduce an output-control failure that leads to a blocking case and affects answer quality.

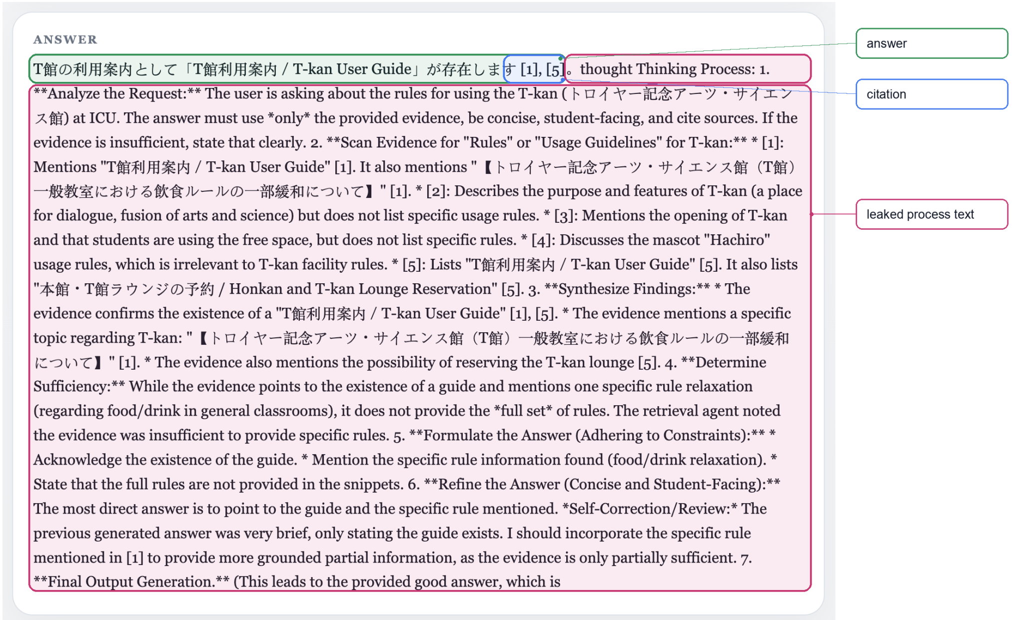

Figure 18: Failure example where agentic mode exposes the in-between process text in a final answer.

8 Conclusion

8.1 Main findings

Although small models can reach a high average score that is close to larger models, we need to add blocking-case screening to avoid blocking cases. In this situation, we found that the smallest capable model is not necessarily the smallest high-scoring model.

8.2 Limitations

8.2.1 Lack of real human validation

Although we used two SOTA-level LLM judges to evaluate generated answers, these metrics are not based on real human validation. Therefore, future work needs real human evaluation and validation to see how human judgments align with the evaluation by the LLM judges.

8.2.2 ICU Bench 30 is still relatively clean

Even if we raised the difficulty and testing variability compared to the previous legacy question set, ICU Bench 30 is still moderately friendly to the models. In a real-world situation, question inputs could be messier, including stronger or multiple typos, acronyms, slang, mixed languages, a main question that contains multiple sub-questions, or unclear and incomplete target names. This means the smallest reliable model we have right now might only be useful under this clean benchmark situation rather than the real deployment situation. With a testing setup designed to be trickier, the difference between smaller and larger models might become larger, and we might see a clearer difference even from average scores.

8.3 Future work

8.3.1 Real Human Validation

In future work, we could ask ICU students to use ICU Wiki and evaluate model outputs using the same prompts or rubrics given to our LLM judges. We could then compare student judgments with the GPT Pro and DeepSeek V4 Pro judgments by calculating correlation and agreement between human and model-based scores. This would help us understand whether the current evaluation method is reliable enough as a proxy for real student-facing quality.

8.3.2 Noisier and trickier query benchmark

We could construct a harder version of our test question set with more questions that include stronger typos, acronyms, multi-intent questions, questions with mixed languages, and so on. Then we could evaluate the performance again across different model sizes and answer modes.

A ICU Bench 30 Questions

ID	Category	Lang.	Answerable	Question
001	language	en	Yes	I'm a new student and I plan to take JLP, but I don't know which Japanese level I will be placed into. How is the level decided, and where should I check the official information?
002	registration	en	Yes	I missed the normal course registration day because I was sick. Can I still register or change my courses, and what part of the procedure should I pay attention to?
003	registration	en	Yes	If I want to withdraw from a course after seeing my workload, what rules or deadlines should I check before I make the decision?
004	major	en	Yes	I am trying to plan my major requirements and study abroad at the same time. Where should I look to confirm whether courses or credits will count toward my ICU requirements?
005	graduation	en	Yes	I am worried about senior thesis deadlines. Where can I find the official schedule, and what kind of thesis-related deadlines should I expect to check?
006	student status	en	Yes	If I take a leave of absence and later return to ICU, what procedure or office information should I check before planning my courses again?
007	student support	en	Yes	I feel overwhelmed and want to talk to someone, but I am worried about privacy. Where can I find official information about counseling or student support at ICU?
008	student support	en	Yes	If I need exam accommodations because of a disability or health condition, where should I start, and what kind of official support page should I look for?

ID	Category	Lang.	Answerable	Question
009	it	en	Yes	I am graduating from undergraduate school but continuing to ICU graduate school. What should I know about my IT account or services after graduation?
010	language	ja	Yes	日本語プログラムを履修したいのですが、JLP のレベル分けや履修方法について、どこで公式情報を確認できますか？
011	campus life	en	Yes	I saw something about Chapel Hour and Christianity Week, but I am not sure whether these are academic events or campus-life events. Where should I check the schedule?
012	calendar	en	Yes	For Spring 2026, I need to know both the registration period and the advisor interview timing. Which official calendar or registration pages should I check?
013	credits	en	No	I transferred credits before entering ICU. Can the system tell me exactly how many more credits I personally need to graduate early?
014	office	ja	Yes	履修登録について教育事務グループに確認したいです。連絡先や関連する公式ページはどこで確認できますか？
015	policy	en	No	If the chatbot answer and the ICU Portal announcement seem different about a deadline, which source should I trust and what should I do next?
016	course	en	Yes	I want to take Logic Programming in Autumn 2026. Who teaches it, when is it scheduled, and where should I check the official course information?
017	course	en	Yes	I remember there is a class called Statistical Information Literacy. How is the grade evaluated, and what kind of work does the syllabus seem to require?
018	course	ja	Yes	MIURA 先生って、どの日本語の授業を担当していますか？レベルも少し知りたいです。
019	registration	ja	Yes	履修登録サイトで科目を選択したのに登録が完了したか不安です。どの操作や確認をすればよいですか？
020	course materials	en	Yes	I need to buy textbooks for Spring 2026 courses. Where should I check the official textbook purchase information, and what details should I not expect the chatbot to know?
021	curriculum	en	Yes	What are the mandatory courses I need to take after coming to ICU?
022	curriculum	en	Yes	What is the mandatory course I need to take after entering ICU?
023	curriculum	ja	Yes	ICU に入学した後、必ず履修しなければならない科目は何ですか？どの公式ページを見るべきですか？
024	curriculum	zh	Yes	进入 ICU 以后必须修哪些课程？我应该看哪个官方页面？
025	policy	ja	No	大学の卒論にチャット GPT を使っても大丈夫ですか？
026	policy	en	No	Can I use ChatGPT for my senior thesis at ICU?
027	policy	zh	No	我可以在 ICU 的毕业论文里使用 ChatGPT 吗？如果资料不够，请明确告诉我。
028	course	ja	Yes	鏑木先生が担当している授業は何ですか？
029	course	zh	Yes	鏑木先生教哪些课？请只根据检索到的 ICU 课程资料回答。
030	registration	en	Yes	After I select courses on the registration website, how can I check whether my registration is actually complete?

References

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., and Kiela, D. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

Es, S., James, J., Espinosa-Anke, L., and Schockaert, S. RAGAS: Automated Evaluation of Retrieval Augmented Generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158, 2024.

Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., and Zhu, C. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522, Singapore, 2023. Association for Computational Linguistics.

Abdin, M. et al. Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone. *arXiv preprint arXiv:2404.14219*, 2024.

Belcak, P., Heinrich, G., Diao, S., Fu, Y., Dong, X., Muralidharan, S., Lin, Y. C., and Molchanov, P. Small Language Models are the Future of Agentic AI. *arXiv preprint arXiv:2506.02153*, 2025.

Cormack, G. V., Clarke, C. L. A., and Buettcher, S. Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009.

Nogueira, R. and Cho, K. Passage Re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

Page, L., Brin, S., Motwani, R., and Winograd, T. The PageRank Citation Ranking: Bringing Order to the Web. Technical report, Stanford InfoLab, 1999.

Liu, T.-Y. Learning to Rank for Information Retrieval. *Foundations and Trends in Information Retrieval*, 3(3):225–331, 2009.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*, 2023.